

First-Order Complexity of Regular Languages

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Logic over Words.

- ▶ Every finite alphabet A induces a logical vocabulary $\sigma_A = \{I_a^1 \mid a \in A\}$.
- ▶ Sentences of **FO**[<] describe properties of words.
- ▶ Variables represent positions in a word.
- ▶ $x < y$ means position x is left of position y .
- ▶ A σ_A -structure $\mathcal{A}$ satisfies $I_a(k)$ if “the k -th letter of $\mathcal{A}$ is a .”¹
- ▶ “Every structure is a word:” any structure is first-order interpretable as a binary string.

¹We are identifying $|A|$ with $\{1, 2, \dots, ||A||\} \subset \mathbb{N}$ via the total order <.

Logic over Words.

- ▶ **Example:** $\forall x(\forall y(x < y \vee x = y) \implies l_0(x))$.
- ▶ “If every position is greater than or equal to x , then the bit at position x is 0.”
- ▶ This describes the language given by the regular expression $0(0 + 1)^*$.

Formulas as Circuits.

- ▶ Let ϕ be a sentence of $\mathbf{FO}[<]$. Can inductively build $O(1)$ -depth, $\text{poly}(n)$ -size circuit which agrees with ϕ on $\{0,1\}^n$.
- ▶ If ϕ is quantifier-free, can perform boolean combinations of $x < y$, $I_0(k)$ in $O(1)$ -size.
- ▶ If $\phi = \exists x(\psi)$, wire n circuits for ψ (with x fixed) to an $\vee$ gate.
- ▶ If $\phi = \forall x(\psi)$, use an $\wedge$ gate.
- ▶ Circuit depth depends only on $\text{rk}(\phi)$. Each quantifier blows up size by a factor of $O(n) \implies \text{poly}(n)$ size. ($\mathbf{AC}^0$ circuit.)

Circuits as Formulas

- ▶ The other direction is harder. E.g., $\text{LEN}(2) \in \mathbf{AC}^0 \setminus \mathbf{FO}[<]$.
- ▶ We will need to make $\mathbf{FO}[<]$ more powerful.
- ▶ A numerical predicate P^a is a subset of $\mathbb{N}^a$.
- ▶ $\mathbf{FO}[\text{Arb}]$ is the union of $\mathbf{FO}[\mathcal{P}]$ for all sets of numerical predicates $\mathcal{P}$.
- ▶ **Theorem** [GL84], [Imm87]: $\mathbf{FO}[\text{Arb}] = \mathbf{AC}^0$.
- ▶ Proof for $\subseteq$ is similar to before. Remains to show $\supseteq$.

Circuits as Formulas

Proof [Imm87]:

- ▶ Let $\mathcal{C} = \{C_0, C_1, C_2, \dots\}$ be an **AC**⁰ circuit family with size $\leq n^k$ and depth d .
- ▶ Suppose $\mathcal{C}$ is in alternating normal form with top gate $\wedge$ and bottom layer $\vee$.
- ▶ Negations are pushed down with De Morgan's law. Treated as layer 0/input layer.
- ▶ For each gate, give each input wire a label using k base- n digits $\bar{z} = z_1 \dots z_k$.

Circuits as Formulas

- ▶ For all $i \in [d]$, let $V_i(m, \bar{z}_1, \dots, \bar{z}_i)$ hold if $\bar{z}_1, \dots, \bar{z}_i$ is a legitimate path from output to layer i in C_m .
- ▶ Let $R(m, \bar{Z}, y, b)$ hold if $\bar{Z}$ is a valid path to an input gate in C_m , and the literal is a_y when $b = 1$ and $\neg a_y$ if $b = 0$.
- ▶ We get the formula

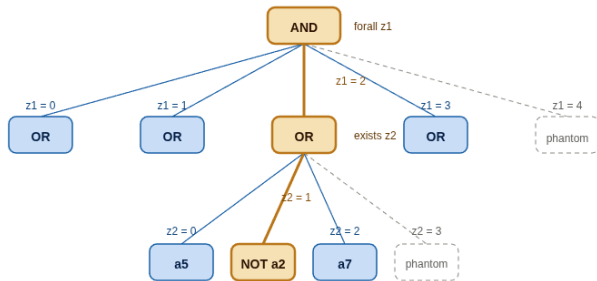
$$\Phi = \forall \bar{z}_1 \exists \bar{z}_2 \forall \bar{z}_3 \dots \exists \bar{z}_d \exists y : (\neg V_1 \vee (V_2 \wedge (\neg V_3 \vee (\dots (\neg V_d \vee \Psi))))$$

where

$$\Psi(\bar{Z}, y) = (R(m, \bar{Z}, y, 1) \wedge I_1(y)) \vee (R(m, \bar{Z}, y, 0) \wedge I_0(y)).$$

Circuits as Formulas

If $w = 1010010$ then $w \models V_1(9, 2) \wedge V_2(9, 2, 1) \wedge R(9, 2, 1, 7, 0)$.
On this path the matrix looks like $0 \vee (1 \wedge (0 \vee 1))$.



What is $\mathbf{FO}[\text{Arb}] \cap \mathbf{REG}$?

- ▶ Schützenberger [Sch65] tells us that $\mathbf{FO}[\prec]$ is generated by regular expressions using complement instead of Kleene star.
- ▶ Can use Ehrenfeucht–Fraïssé games to show $\mathbf{FO}[\prec] \subsetneq \mathbf{REG}$.
- ▶ $\mathbf{FO}[\mathcal{P}] \subseteq \mathbf{REG}$ if predicates in $\mathcal{P}$ are “computed by a DFA.”
- ▶ Ajtai [Ajt83] and Furst *et. al.* [FSS84] tell us that $\text{PARITY} \notin \mathbf{AC}^0 = \mathbf{FO}[\text{Arb}]$.
- ▶ So what is $\mathbf{FO}[\text{Arb}] \cap \mathbf{REG}$? Can't be all of $\mathbf{REG}$, must be at least star-free...

Monoids and Languages.

- ▶ A *monoid* is a set M with an associative binary operation $\cdot$ and an identity element $1_M \in M$.
- ▶ **Example:** Σ^* is a monoid under concatenation.
- ▶ For $L \subseteq \Sigma^*$, define the *syntactic congruence* $\equiv_L$:

$$x \equiv_L y \iff \forall u, v : uxv \in L \iff uyv \in L.$$

- ▶ The quotient $\Sigma^* / \equiv_L$ is the *syntactic monoid* $\mathcal{M}(L)$ of L .
- ▶ If $L \in \mathbf{REG}$ and M is a minimal DFA, then $\mathcal{M}(L)$ is isomorphic to the *transition monoid*

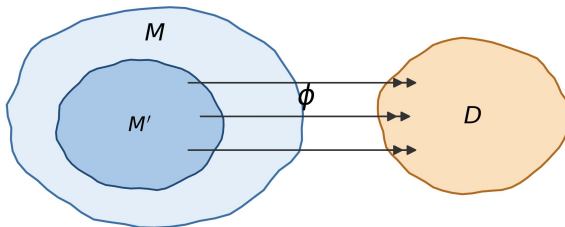
$$\mathcal{T}(M) = \{q \mapsto \delta(q, w) \mid w \in \Sigma^*\}.$$

What is $\mathbf{FO}[\text{Arb}] \cap \mathbf{REG}$?

- ▶ Chandra *et. al.* [CFL85]: A monoid is in $\mathbf{AC}^0$ iff it contains no non-trivial group. We don't know about regular languages though...
- ▶ **Lower bound idea #1:** If $\mathcal{M}(L')$ contains $\mathcal{M}(L)$, then $L \leq L'$.
- ▶ But this fails: $\mathcal{M}(\text{LEN}(4)) = \mathbb{Z}/4\mathbb{Z}$ but $\mathcal{M}(\text{PARITY}) = \mathbb{Z}/2\mathbb{Z}$.
- ▶ **Lower bound idea #2:** If $\eta_{L'}$ can be computed using η_L and only a little extra computation, then $L \leq L'$.

Monoids and Languages.

- A monoid D divides a monoid M if there exists a subset $M' \subseteq M$ and a surjective monoid homomorphism $\phi : M' \rightarrow D$.



Monoids and Languages.

- ▶ Let M be a monoid and $\phi : \Sigma^* \rightarrow M$ be a homomorphism.
- ▶ A language $L \subseteq \Sigma^*$ is *recognized by* ϕ if there exists a subset $T \subseteq M$ such that

$$w \in L \iff \phi(w) \in T.$$

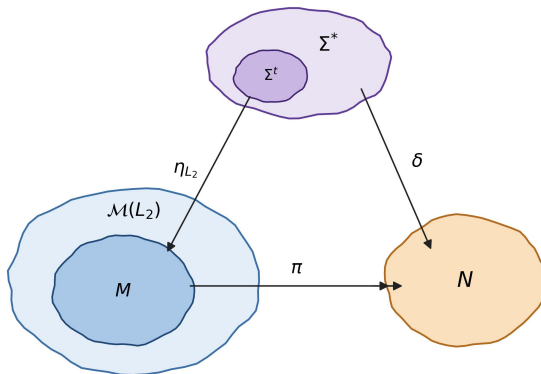
- ▶ Every $L \in \mathbf{REG}$ is recognized by η_L . Take $M = \mathcal{M}(L)$ and $T = \{[w]_{\equiv_L} \mid w \in L\}$.

Regular Language Reduction Lemma

Lemma:

- ▶ Let $L_1, L_2 \in \mathbf{REG}$.
- ▶ Suppose L_1 is recognized by a homomorphism $\delta : \Sigma^* \rightarrow N$ where N divides $\mathcal{M}(L_2)$ via $\pi : M \twoheadrightarrow N$ for some $M \subseteq \mathcal{M}(L_2)$.
- ▶ Suppose also that there exists $t > 0$ such that for all $a \in \Sigma$, there exists $w \in \Sigma^t$ where $\pi(\eta_{L_2}(w)) = \delta(a)$.
- ▶ Then $L_1 \leq_{\mathbf{NC}^0}^T L_2$.

Regular Language Reduction Lemma



$$\forall a \in \Sigma \exists w \in \Sigma^t: \pi(\eta_{L_2}(w)) = \delta(a)$$

Regular Language Reduction Lemma

Proof [BCST92]:

- ▶ For all $u, v \in \Sigma^*$, the (u, v) -quotient of L_2 is

$$u^{-1}L_2v^{-1} := \{w \in \Sigma^* \mid uwv \in L_2\}.$$

- ▶ If $w \equiv_{L_2} w'$, then $w \in u^{-1}L_2v^{-1} \iff w' \in u^{-1}L_2v^{-1}$.
- ▶ So (u, v) -quotients are unions of $\equiv_{L_2}$ -classes.
- ▶ We assumed $L_2 \in \mathbf{REG}$, so $\mathcal{M}(L_2)$ is finite.
- ▶ There must be only finitely many (u, v) -quotients.

Regular Language Reduction Lemma

- ▶ Pick $u_1, v_1, \dots, u_k, v_k \in \Sigma^*$ to exhaust the (u, v) -quotients of L_2 .
- ▶ Define $\rho : \Sigma^* \rightarrow \{0, 1\}^k$ by

$$\rho(w) = (\chi_{u_1^{-1}L_2v_1^{-1}}(w), \dots, \chi_{u_k^{-1}L_2v_k^{-1}}(w)).$$

- ▶ **Claim 1:** ρ tells us all we need to know to determine $\eta_{L_2}(w)$.
- ▶ We will show there exists a finite function $\phi : \{0, 1\}^k \rightarrow \mathcal{M}(L_2)$ such that $\eta_{L_2} = \phi \circ \rho$.

Regular Language Reduction Lemma

- ▶ Suppose $\rho(w) = (b_1, \dots, b_k)$.
- ▶ Let $\text{Good} = \{u_i^{-1}L_2v_i^{-1} \mid b_i = 1\}$.
- ▶ **Claim 2:**

$$\eta_{L_2}(w) = \bigcap_{Q \in \text{Good}} Q$$

- ▶ Suffices to show no $w' \notin \eta_{L_2}(w)$ is in the intersection.
- ▶ For all $w' \not\equiv_{L_2} w$, there exists $u, v \in \Sigma^*$ such that $uwv \in L_2$ but $uw'v \notin L_2$.
- ▶ Then $u^{-1}L_2v^{-1}$ “deletes” w' :

$$w \in u^{-1}L_2v^{-1} \implies u^{-1}L_2v^{-1} \in \text{Good} \implies w' \notin \bigcap_{Q \in \text{Good}} Q$$

Regular Language Reduction Lemma

► So

$$\phi(b_1, \dots, b_k) = \bigcap_{Q \in \text{Good}} Q$$

is a well-defined map $\{0, 1\}^k \rightarrow \mathcal{M}(L_2)$.

- Since $\phi(\rho(w)) = \eta_{L_2}(w)$ we have proven Claim 1.
- Finally, we construct an $\mathbf{NC}^0[L_2]$ circuit for L_1 .

Regular Language Reduction Lemma

Recall: for all $a \in \Sigma$, there exists $w_a \in \Sigma^t$ s.t. $\pi(\eta_{L_2}(w_a)) = \delta(a)$.

1. Compute $x_i \mapsto w_{x_i}$ for all $i \in [2, n-1]$.
2. Compute $x_1 \mapsto u_i w_{x_1}$ and $x_n \mapsto w_{x_n} v_i$ for all $i \in [k]$.
► **Important:** 1 and 2 require $|w_a| = t$ assumption.
3. Compute $\chi_{L_2}(u_i w_{x_1} \dots w_{x_n} v_i)$ for all $i \in [k]$ with L_2 gates.
This yields $\rho(w_{x_1} \dots w_{x_n})$.
4. Compute ϕ on $\rho(w_{x_1} \dots w_{x_n})$ to get $\lceil \log_2 |\mathcal{M}(L_2)| \rceil$ bits representing $\eta_{L_2}(w_{x_1} \dots w_{x_n})$.
5. Compute π on η_{L_2} to get $\lceil \log_2 |N| \rceil$ bits representing an element of N . Check if it is in the accepting subset.

Regular Language Reduction Lemma

Sanity Check:

$$\begin{aligned}\pi(\eta_{L_2}(w_{x_1} \dots w_{x_n})) &= \pi(\eta_{L_2}(w_{x_1}) \dots \eta_{L_2}(w_{x_n})) \\ &= \pi(\eta_{L_2}(w_{x_1})) \dots \pi(\eta_{L_2}(w_{x_n})) \\ &= \delta(x_1) \dots \delta(x_n).\end{aligned}$$

- ▶ Each step uses $O(1)$ depth, $O(n)$ gates, $O(1)$ fan-in (except L_2 gates) $\implies L_1 \leq_{\mathbf{NC}^0}^t L_2$.
- ▶ This is surely a “first-order Turing reduction,” but no one has bothered showing it.

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

- ▶ Let $L \in \mathbf{REG}$.
- ▶ **Definition:** $\mathcal{D}(L)$ is the set of monoids N such that there exists $t > 0$ where N divides $\eta_L(\Sigma^t)$ via a homomorphism π .
- ▶ If $N \in \mathcal{D}(L)$ then $N \leq_{\mathbf{NC}^0}^T L$ by RLRL.
- ▶ **Theorem** ([BCST92]): If $L \in \mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$, then $\mathcal{D}(L)$ contains no non-trivial group.

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

Proof:

- ▶ Suppose $L \in \mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$ and $G \in \mathcal{D}(L)$ is a non-trivial group.
- ▶ Then $\eta_L(\Sigma^t)$ must contain a non-trivial cyclic group $M = \langle m \rangle$. (Take π^{-1} of such a group in G .)
- ▶ Pick $w_0, w_1 \in \Sigma^t$ such that $\eta_L(w_0) = 1_M$ and $\eta_L(w_1) = m$.
- ▶ Define a homomorphism $\delta : \{0, 1\}^* \rightarrow M$ by $\delta(0) = 1_M$ and $\delta(1) = m$. This recognizes $\text{MOD}(k)$.
- ▶ Note $\eta_L(w_0) = \delta(0)$ and $\eta_L(w_1) = \delta(1)$, so RLRL applies.
- ▶ Combine with Smolensky [Smo87] $\text{MOD}(k) \notin \mathbf{AC}^0$:

$$\text{MOD}(k) \leq_{\mathbf{NC}^0}^t L \implies L \notin \mathbf{FO}[\mathbf{Arb}]$$

What is $\mathbf{FO[Arb]} \cap \mathbf{REG}$?

- ▶ What if $\mathcal{D}(L)$ contains no non-trivial group?
- ▶ It turns out we only need predicates which are themselves “regular.”

Regular Numerical Predicates

- ▶ Let $\mathcal{V}$ be a set of variables.
- ▶ **Definition:** A $\mathcal{V}$ -structure over Σ is a word

$$(a_1, U_1)(a_2, U_2) \dots (a_r, U_r) \in (\Sigma \times 2^{\mathcal{V}})^*$$

where $U_i \cap U_j = \emptyset$ and $\bigcup_i U_i = \mathcal{V}$.

- ▶ A $\mathcal{V}$ -structure $\mathbf{w}$ is *suitable* for $\phi(\mathcal{V})$ if bound variables do not appear in $\mathcal{V}$ and each quantifier binds a distinct variable.
- ▶ Variables take the values of the positions they appear in.

Regular Numerical Predicates

- ▶ Fix a unary alphabet $\Gamma = \{1\}$ and take any numerical predicate $R \subseteq \mathbb{N}^k$.
- ▶ A finite automaton M over $\Gamma \times 2^{\{x_1, \dots, x_k\}}$ computes R if M accepts precisely the $\{x_1, \dots, x_k\}$ -structures with prefixes

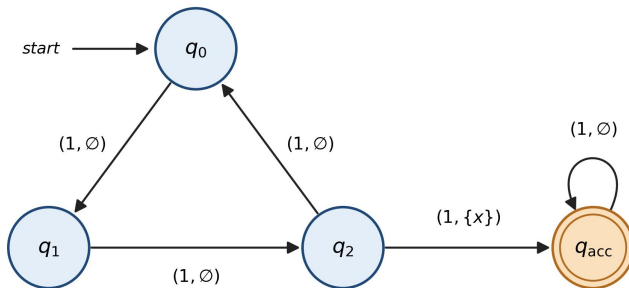
$$(1, \emptyset)^{n_1-1}(1, \{x_1\})(1, \emptyset)^{n_2-1}(1, \{x_2\}) \dots (1, \emptyset)^{n_k-1}(1, \{x_k\})$$

for all $(n_1, \dots, n_k) \in R$.

- ▶ **Definition:** A k -ary numerical predicate is *regular* iff it is computed by a DFA over $\Gamma \times 2^{\{x_1, \dots, x_k\}}$.

Regular Numerical Predicates

Example: $\text{MOD}_3(x)$ is a regular predicate.



Regular Numerical Predicates

- ▶ **Theorem** [Str94]: The set of regular predicates **Reg** contains only MOD_m for all $m > 0$, and $x < y$.
- ▶ The proof relies on Büchi's result that **REG** = **MSO**[+1] [Bü60]. It is rather tedious so we will move on with our lives.

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

- ▶ **Theorem** [Str94]: If $L \in \mathbf{REG}$ and $\mathcal{D}(L)$ contains no non-trivial group, then $L \in \mathbf{FO}[\mathbf{Reg}]$.
- ▶ **Corollary:** $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG} = \mathbf{FO}[\mathbf{Reg}]$.

$$\begin{aligned} L \in \mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG} &\implies \mathcal{D}(L) \text{ contains no non-trivial group.} \\ &\implies L \in \mathbf{FO}[\mathbf{Reg}] \\ &\implies L \in \mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG} \end{aligned}$$

Relativization Lemma

Lemma:

- ▶ Let ϕ be an **FO**[Arb] sentence.
- ▶ Let w be an $\{x\}$ -structure over Σ .
- ▶ Let $w' \in \Sigma^*$ be the prefix of w appearing before x .
- ▶ Then there exists an **FO**[Arb] formula $\phi[< x]$ such that

$$w \models \phi[< x] \iff w' \models \phi.$$

Relativization Lemma

Proof (structural induction):

- ▶ If ϕ is an atomic formula then $\phi[< x] = \phi$.
- ▶ If $\phi = \alpha \wedge \beta$ then $\phi[< x] = \alpha[< x] \wedge \beta[< x]$.
- ▶ If $\phi = \neg\alpha$ then $\phi[< x] = \neg(\alpha[< x])$.
 - ▶ **Easy exercise:** Verify the above claims.
- ▶ If $\phi = \exists y(\alpha)$ then $\phi[< x] = \exists y((y < x) \wedge \alpha[< x])$.

Relativization Lemma

$$\phi = \exists y(\alpha) \quad \longleftarrow \quad \phi[< x] = \exists y((y < x) \wedge \alpha[< x]).$$

- ▶ Suppose x occurs at position k in w and $w \models \phi[< x]$.
- ▶ Pick witness $j < k$ for y ; stick y in position j of w .
- ▶ This yields w_y such that $w_y \models \alpha[< x]$.
- ▶ By induction, α holds in the $(k - 1)$ -length prefix w'_y of w_y .
- ▶ Take y back out: $w' \models \exists y(\alpha)$.

Relativization Lemma

$$\phi = \exists y(\alpha) \quad \longrightarrow \quad \phi[< x] = \exists y((y < x) \wedge \alpha[< x]).$$

- ▶ Suppose $w' \models \exists y(\alpha)$.
- ▶ Pick a witness $j \leq |w'| = k - 1$ for y ; stick y in position j of w' .
- ▶ This yields w'_y such that $w'_y \models \alpha$.
- ▶ Let s be the $(|w| - k + 1)$ -length suffix of w . By induction,

$$w'_y s \models \alpha[< x].$$

- ▶ Take y back out: $w = w'_y s \models \exists y((y < x) \wedge \alpha[< x])$.

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

Now we will prove:

- ▶ **Theorem:** If $L \in \mathbf{REG}$ and $\mathcal{D}(L)$ contains no non-trivial group, then $L \in \mathbf{FO}[\mathbf{Reg}]$.

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

Proof:

- ▶ Suppose $\mathcal{D}(L)$ (hence $\eta_L(\Sigma^t)$'s) contains no non-trivial group.
- ▶ There are finitely many sets $\eta_L(\Sigma), \eta_L(\Sigma^2), \eta_L(\Sigma^3), \dots$.
- ▶ Exists $k, r > 0$ such that $\eta_L(\Sigma^p) = \eta_L(\Sigma^{p+r})$ for all $p \geq k$.
- ▶ Pick p divisible by r , whence $S = \eta_L(\Sigma^p) = \eta_L((\Sigma^p)^+)$.
- ▶ S is almost a monoid, just missing 1.

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

- ▶ For all $w \in \Sigma^*$, we consider $Lw^{-1} \cap \mathbf{LEN}(p)$ which we denote

$$L_w = \{u \in (\Sigma^p)^* \mid uw \in L\}.$$

- ▶ Observe:

$$L = \bigcup_{w \in \Sigma^{< p}} L_w w.$$

- ▶ Suffices to show each $L_w w \in \mathbf{FO}[\mathbf{Reg}]$.
- ▶ In fact, we just need each $L_w \in \mathbf{FO}[\mathbf{Reg}]$. Consider any $w = a_1 \dots a_\ell$ for $\ell < p$.

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

- ▶ If $\phi \in \mathbf{FO}[\mathbf{Reg}]$ expresses L_w , relativization yields $\mathbf{FO}[\mathbf{Reg}]$ sentence for $L_w w$:

$$\begin{aligned} \exists x \exists y_1 \dots \exists y_\ell (\phi[< x] \wedge \text{Suc}(x, y_1) \\ \wedge \left(\bigwedge_{i=1}^{\ell-1} \text{Suc}(y_i, y_{i+1}) \wedge I_{a_i}(y_i) \right) \wedge I_{a_\ell}(y_\ell)). \end{aligned}$$

- ▶ “Everything $< x$ is an L_w string and everything $\geq x$ is just w .”

What is $\mathbf{FO}[\text{Arb}] \cap \mathbf{REG}$?

- ▶ Let $B = \Sigma^P$ be a new alphabet and define $\beta : B \rightarrow S$ by

$$\beta(b) = \eta_L(b).$$

- ▶ Extend β to a homomorphism $B^* \rightarrow S \cup \{1_{\mathcal{M}(L)}\}$.
- ▶ A word $u \in B^*$ as an element of $(\Sigma^P)^*$ is in L_w iff

$$\beta(u) \in T = \{m \in S \cup \{1_{\mathcal{M}(L)}\} \mid m \cdot \eta_L(w) \in \eta_L^{-1}(L)\}.$$

- ▶ β **recognizes L_w as a subset of B^*** . Moreover, $S \cup \{1_{\mathcal{M}(L)}\}$ contains no non-trivial group.
- ▶ Any language recognized by such a monoid cannot have a non-trivial group in its syntactic monoid.
- ▶ Schützenberger's theorem therefore furnishes an $\mathbf{FO}[<]$ formula ψ for L_w with B as alphabet! Beautiful!

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

- ▶ We need to translate the $\mathbf{FO}[<]$ sentence ψ over alphabet B into an $\mathbf{FO}[\mathbf{Reg}]$ sentence ϕ over alphabet Σ .
- ▶ **Algorithm:** Starting with outside quantifiers, replace $\exists x(\alpha)$ with

$$\exists x(\text{MOD}_p(x) \wedge \alpha).$$

- ▶ Replace each $I_b(x)$ where $b \in B$ is $a_1 \dots a_p \in \Sigma^p$ with

$$I_{a_p}(x) \wedge \exists y_1 \dots \exists y_p((y_p = x) \wedge \left(\bigwedge_{i=1}^{p-1} \text{Suc}(y_i, y_{i+1}) \wedge I_{a_i}(y_i) \right) \wedge I_{a_p}(y_p))$$

- ▶ “Unfolding $b \in B$ at position x from right to left.”

What is $\mathbf{FO}[\mathbf{Arb}] \cap \mathbf{REG}$?

- ▶ Take result of algorithm and conjunct with

$$\forall x \forall y ((y \leq x) \implies \text{MOD}_p(y))$$

to weed out strings not in $(\Sigma^p)^*$.

- ▶ We now have an $\mathbf{FO}[\mathbf{Reg}]$ formula for L_w . Conclusion:
 $\mathcal{D}(L)$ contains no non-trivial group $\implies L \in \mathbf{FO}[\mathbf{Reg}]$.

“McNaughton’s Last Conjecture”

McNaughton’s 1960 Conjecture:

$$\mathbf{FO[Arb]} \cap \mathbf{REG} = \mathbf{FO[Reg]}$$

“I regret that the reasons for my conjectures... are too complicated, and by themselves inconsequential, for presentation here.”



Source: [Str18]

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